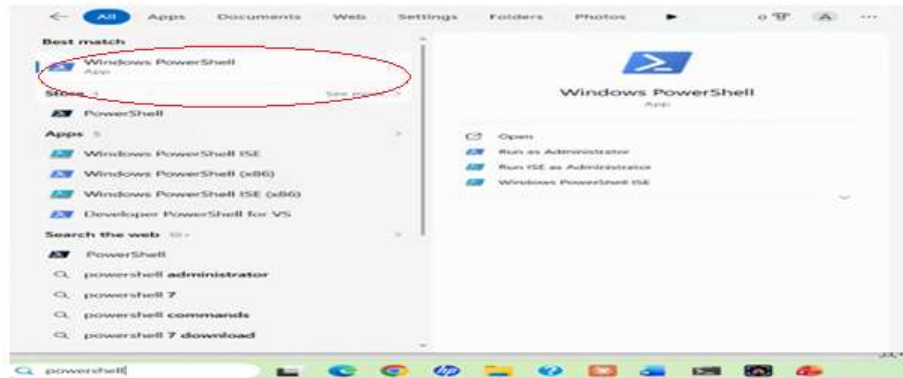
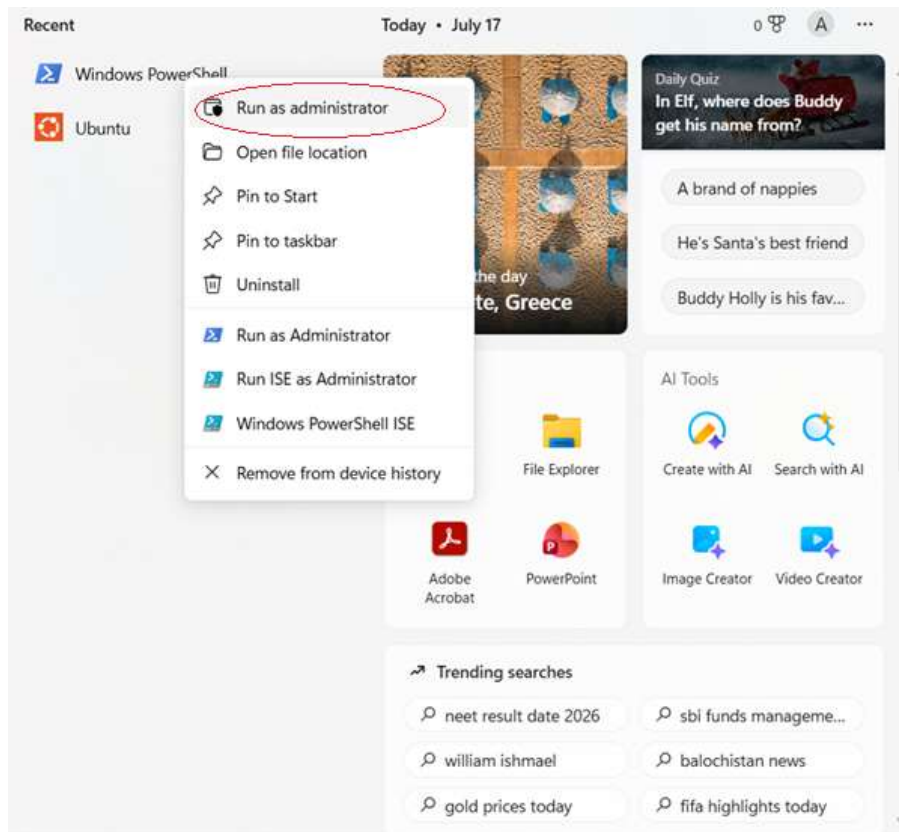


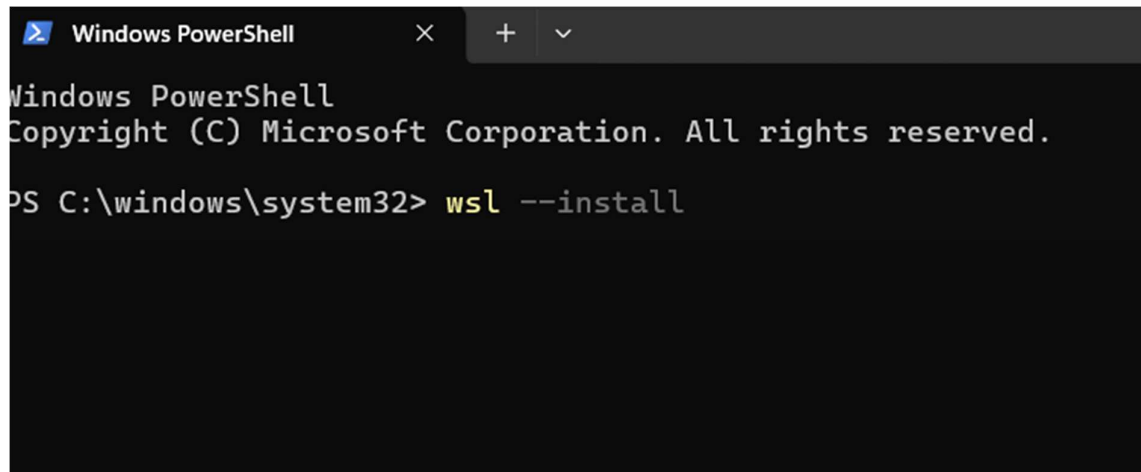
Step 1: Open PowerShell as Administrator

1. Press **Windows**.
2. Search for **PowerShell**.



3. Right-click **Windows PowerShell**.



A screenshot of a Windows PowerShell terminal window. The title bar shows 'Windows PowerShell' with a close button and window controls. The terminal text reads: 'Windows PowerShell', 'Copyright (C) Microsoft Corporation. All rights reserved.', and 'PS C:\windows\system32> wsl --install'.

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\windows\system32> wsl --install
```

Step 3: Restart Your Computer

After installation, restart your PC when prompted.

Step 4: Complete Ubuntu Setup

After restarting:

1. Ubuntu opens automatically (or search for **Ubuntu** in the Start menu).
2. Wait for the installation to finish.
3. Create a Linux username.

Example:

Enter new UNIX username: anusha

Step 5: Create a Password

You'll then be prompted to create a password.

New password: ****

Retype new password:****

Step 6: Update Ubuntu

```

nikhilreddy@Hp:/mnt/c/windows/system32$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:3 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [417 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [97.2 kB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.7 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [159 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [51.2 kB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.2 kB]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [349 kB]
Get:11 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [67.8 kB]
Hit:12 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:13 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [492 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [118 kB]
Get:15 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [97.8 kB]
Get:16 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [243 kB]
Get:17 http://archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [81.0 kB]
Get:18 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [184 kB]
Get:19 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [367 kB]
Get:20 http://archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [70.8 kB]
Fetched 3159 kB in 2s (1740 kB/s)
123 packages can be upgraded. Run 'apt list --upgradable' to see them.

```

Step 7: Install GCC (C Compiler)

```

nikhilreddy@Hp:/mnt/c/windows/system32$ sudo apt install build-essential
Upgrading:
  build-essential

Installing dependencies:
  libcrypt-dev

Summary:
  Upgrading: 1, Installing: 1, Removing: 0, Not Upgrading: 122
  Download size: 127 kB
  Space needed: 384 kB / 1024 GB available

Continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libcrypt-dev amd64 1:4.5.1-1 [122 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 build-essential amd64 12.12ubuntu2.26.04.2 [5158 B]
Fetched 127 kB in 0s (522 kB/s)
Selecting previously unselected package libcrypt-dev:amd64.
(Reading database ... 42263 files and directories currently installed.)
Preparing to unpack .../libcrypt-dev_1%3a4.5.1-1_amd64.deb ...
Unpacking libcrypt-dev:amd64 (1:4.5.1-1) ...
Preparing to unpack .../build-essential_12.12ubuntu2.26.04.2_amd64.deb ...
Unpacking build-essential (12.12ubuntu2.26.04.2) over (12.12ubuntu2) ...
Setting up libcrypt-dev:amd64 (1:4.5.1-1) ...
Setting up build-essential (12.12ubuntu2.26.04.2) ...
Processing triggers for man-db (2.13.1-1build1) ...

```

Step 8: Verify the Installation

```

nikhilreddy@Hp:/mnt/c/windows/system32$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

```

Step 9: Create Your C Program

Example: nano process.c

Step 10: Compile the Program

```
nikhilreddy@Hp:/mnt/c/windows/system32$ nano process.c
nikhilreddy@Hp:/mnt/c/windows/system32$ gcc process.c -o process
nikhilreddy@Hp:/mnt/c/windows/system32$ ./process
Parent Process
Parent PID : 904
Child PID  : 905

Child Process
Child PID : 905
Parent PID: 904

Enter a Linux command: pwd
/mnt/c/windows/system32
Child process completed.
```

nikhilreddy@Hp: /mnt/c/windows/system32

GNU nano 8.7.1

```
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <stdlib.h>

int main()
{
    pid_t pid;
    char command[100];

    printf("Parent Process\n");

    pid = fork();

    if (pid < 0)
    {
        perror("Fork failed");
        return 1;
    }

    if (pid > 0)
    {
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n\n", pid);

        wait(NULL);

        printf("Child process completed.\n");
    }
    else
    {
        printf("Child Process\n");
        printf("Child PID : %d\n", getpid());
        printf("Parent PID: %d\n\n", getppid());

        printf("Enter a Linux command: ");
        scanf("%99s", command);

        system(command);
    }

    return 0;
}
```

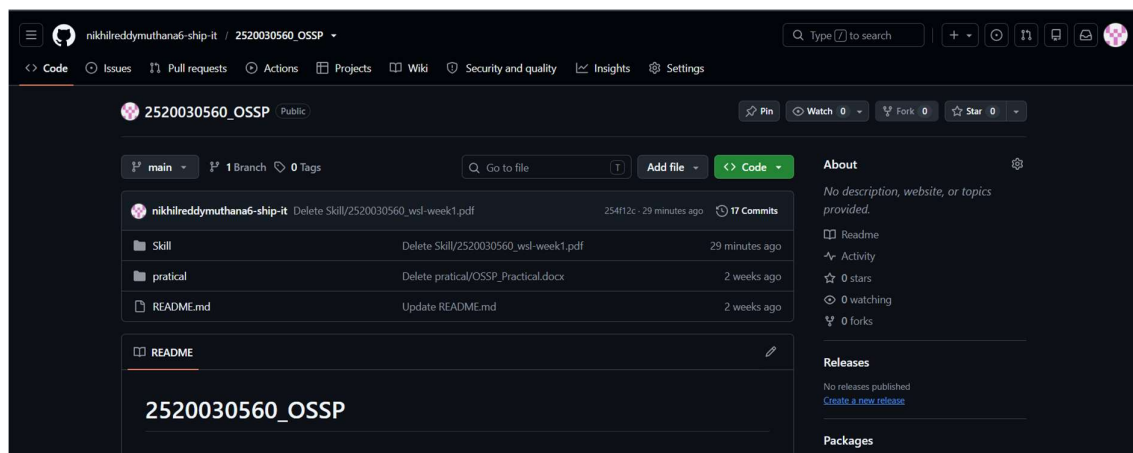
```
nikhilreddy@Hp: ~  
GNU nano 8.7.1  
#include <stdio.h>  
#include <unistd.h>  
  
int main()  
{  
    printf("Before exec()\n");  
  
    execlp("ls", "ls", "-l", NULL);  
  
    perror("exec failed");  
  
    return 1;  
}
```

```
nikhilreddy@Hp:~$ nano exec.c  
nikhilreddy@Hp:~$ gcc exec.c -o exec  
nikhilreddy@Hp:~$ ./exec  
Before exec()  
total 200  
-rwxr-xr-x 1 nikhilreddy nikhilreddy 16088 Aug 14 09:47 childprocess  
-rw-r--r-- 1 nikhilreddy nikhilreddy 252 Aug 14 09:47 childprocess.c  
-rw-r--r-- 1 nikhilreddy nikhilreddy 0 Jul 31 04:02 example.txt  
-rwxr-xr-x 1 nikhilreddy nikhilreddy 15992 Aug 14 10:14 exclp  
-rw-r--r-- 1 nikhilreddy nikhilreddy 178 Aug 14 10:14 exclp.c  
-rwxr-xr-x 1 nikhilreddy nikhilreddy 16040 Aug 24 05:30 exec  
-rw-r--r-- 1 nikhilreddy nikhilreddy 166 Aug 24 05:30 exec.c  
-rw-r--r-- 1 nikhilreddy nikhilreddy 78 Jul 31 09:36 filename.c  
-rw-r--r-- 1 nikhilreddy nikhilreddy 534 Aug 11 13:31 input.txt  
-rw-r--r-- 1 nikhilreddy nikhilreddy 186 Aug 14 10:06 multiplechildprocesses  
-rw-r--r-- 1 nikhilreddy nikhilreddy 186 Aug 14 10:07 multiplechildprocesses.c  
-rw-r--r-- 1 nikhilreddy nikhilreddy 0 Aug 14 10:08 mutiplechildprocesses.c  
-rwxr-xr-x 1 nikhilreddy nikhilreddy 16168 Aug 11 13:31 nikhil  
-rw-r--r-- 1 nikhilreddy nikhilreddy 566 Aug 11 03:59 nikhil.c  
-rwxr-xr-x 1 nikhilreddy nikhilreddy 16168 Aug 11 13:20 nikhil1  
-rw-r--r-- 1 nikhilreddy nikhilreddy 534 Aug 11 13:22 nikhil1.c  
-rwxr-xr-x 1 nikhilreddy nikhilreddy 15952 Jul 31 09:36 nikhilprograms  
drwxr-xr-x 2 nikhilreddy nikhilreddy 4096 Jul 28 04:07 nikhilreddy  
-rw-r--r-- 1 nikhilreddy nikhilreddy 534 Aug 11 13:31 output.txt  
-rwxr-xr-x 1 nikhilreddy nikhilreddy 16088 Aug 14 10:15 parentandchild  
-rw-r--r-- 1 nikhilreddy nikhilreddy 335 Aug 14 10:15 parentandchild.c  
-rwxr-xr-x 1 nikhilreddy nikhilreddy 16088 Aug 14 09:57 parentwaitsforchild  
-rw-r--r-- 1 nikhilreddy nikhilreddy 274 Aug 14 09:57 parentwaitsforchild.c  
-rwxr-xr-x 1 nikhilreddy nikhilreddy 16000 Aug 14 09:38 process  
-rw-r--r-- 1 nikhilreddy nikhilreddy 119 Aug 14 09:38 process.c
```



```
nikhilreddy@Hp: ~  
GNU nano 8.7.1  
#include <stdio.h>  
#include <unistd.h>  
#include <sys/types.h>  
  
int main()  
{  
    pid_t pid;  
  
    printf("Before fork()\n");  
  
    pid = fork();  
  
    if (pid < 0)  
    {  
        perror("fork failed");  
        return 1;  
    }  
    else if (pid == 0)  
    {  
        printf("Child Process\n");  
        printf("Child PID : %d\n", getpid());  
        printf("Parent PID : %d\n", getppid());  
    }  
    else  
    {  
        printf("Parent Process\n");  
        printf("Parent PID : %d\n", getpid());  
        printf("Child PID : %d\n", pid);  
    }  
  
    return 0;  
}
```

```
nikhilreddy@Hp:~$ nano exec.c
nikhilreddy@Hp:~$ nano fork.c
nikhilreddy@Hp:~$ gcc fork.c -o fork
nikhilreddy@Hp:~$ ./fork
Before fork()
Parent Process
Parent PID : 13667
Child PID : 13668
Child Process
Child PID : 13668
Parent PID : 13667
nikhilreddy@Hp:~$
```



https://github.com/nikhilreddymuthana6-ship-it/2520030560_OSSP